

FORM TP 2012177



TEST CODE **02120020**

MAY/JUNE 2012

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

ECOLOGY, HUMAN POPULATION AND NATURAL RESOURCES

UNIT 1 – Paper 02

2 hours 30 minutes

03 MAY 2012 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Do **NOT** open this examination paper until instructed to do so.
2. This paper consists of **SIX** questions, **TWO** from each Module.
3. Candidates **MUST** answer **ALL** questions.
4. Write your answers in the answer booklet provided.
5. You may use a silent, non-programmable calculator to answer items.

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02120020/CARE 2012

MODULE 1

Answer BOTH questions.

1. Figure 1 shows the population size of the aquatic duckweed, *Lemna minor*. This weed grows in canals adjacent to agricultural farms that discharge effluents enriched with agrochemicals, one of which is a limiting factor for the growth of the duckweed. Study Figure 1 and answer the questions that follow.

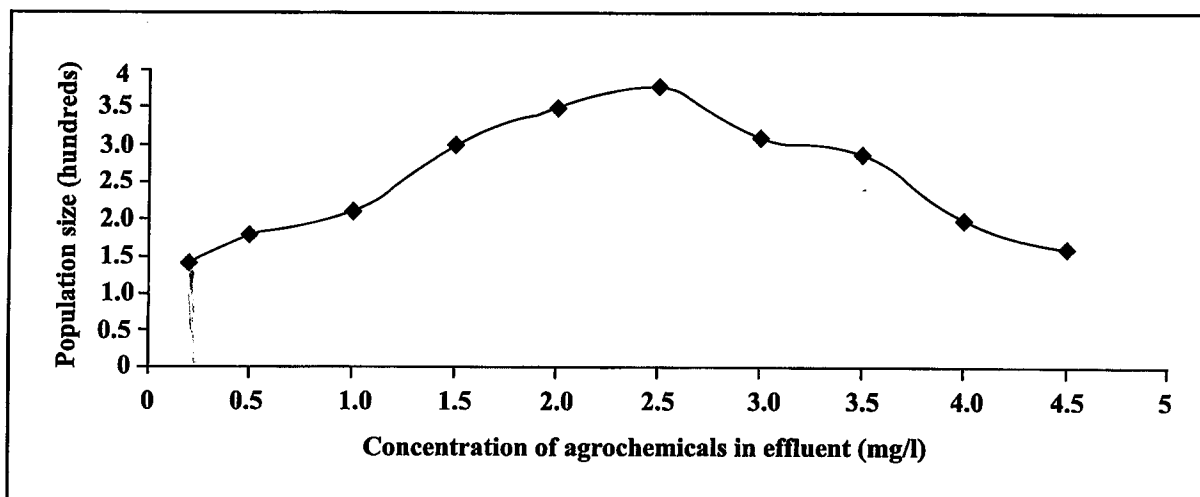


Figure 1. Variation in population size of duckweed

- (a) Using an example, other than agrochemicals, explain what is meant by the term 'limiting factor'. [4 marks]
- (b) Using the information provided in Figure 1, describe the trend in population size of the duckweed. [4 marks]
- (c) It was determined that there was a high concentration of phosphate and nitrate in the effluent. Using the information in Figure 1, explain
- (i) why the effluent concentration resulted in the trend observed in (b) [8 marks]
 - (ii) how the variation in the duckweed population may impact on the population of other aquatic species in the same ecosystem. [4 marks]

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Total 20 marks

2. (a) Distinguish between 'gaseous biogeochemical cycles' and 'sedimentary biogeochemical cycles'. In your response give ONE example of a gaseous biogeochemical cycle and ONE example of a sedimentary biogeochemical cycle. [4 marks]
- (b) "In a food chain there is a unidirectional flow of energy from the sun to producers and then to a series of consumers of various types."
- (i) State the ten per cent rule for the transfer of energy from one trophic level to the next in an ecosystem. [2 marks]
- (ii) On the basis of the rule stated in (b) (i), explain why it will be more energy efficient for humans to eat corn rather than beef. [3 marks]
- (c) Figure 2 shows the current forest area as a percentage of the original forest cover in three Caribbean countries, Guyana, Haiti and Dominica.

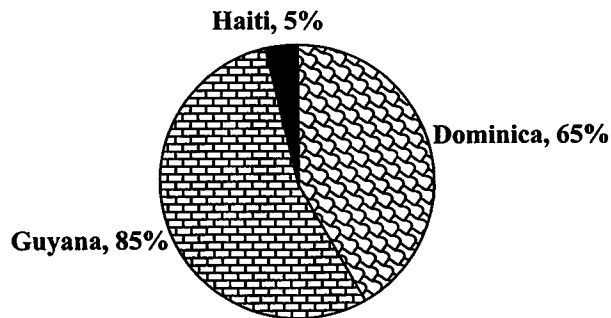


Figure 2. Intact forest as a percentage of original forest cover

- (i) Showing how you arrived at your answer, determine which TWO countries together have a combined total of 50% deforestation. [4 marks]
- (ii) Based ONLY on the information presented in Figure 2, which country would most likely have experienced the **greatest** loss of biodiversity? [1 mark]
- (iii) Identify the country where secondary ecological succession is **LEAST** likely to occur. [1 mark]
- (iv) Justify your response in (c) (iii) above. [5 marks]

Total 20 marks

MODULE 2

Answer BOTH questions.

3. Table 1 presents data for per capita CO₂ emissions for two countries, A and B. Study the data and answer the questions that follow.

TABLE 1: PER CAPITA CO₂ EMISSIONS FOR TWO COUNTRIES

Country	Population Size (millions)	Per Capita CO ₂ Emissions (kg)
A	8.3	8580
B	71.5	1465

- (a) Define 'per capita CO₂ emissions'. [3 marks]
- (b) (i) Calculate the total per capita CO₂ emissions for BOTH countries. [4 marks]
- (ii) Based on the results of the calculation in (b) (i), in which of the two countries will the CO₂ emissions most likely have a GREATER impact on the environment? [1 mark]

- ~~(c) "When assessing the relationship between people and their environment, consideration must be given to abiotic and biotic factors."~~

With reference to this statement, explain how EACH of the following can affect the distribution and activities of human populations.

- (i) Extremes of temperature [4 marks]
- (ii) Fertile soil [4 marks]
- (d) Discuss TWO reasons why some people believe that poverty tends to encourage population growth. [4 marks]

Total 20 marks

4. Table 2 presents data on environmental impacts of over-consumption of natural resources in Country A and Country B. Study the data and answer the questions that follow.

**TABLE 2: ENVIRONMENTAL IMPACTS OF OVER-CONSUMPTION
IN TWO COUNTRIES**

Environmental Impact of Over-consumption	Intensity of Impact	
	Country A	Country B
Deforestation	Medium	Medium
Introduction of exotic species	High	No impact
Pollution	Low	High

- (a) (i) In which country will there be a **greater** impact on natural habitats? Give a reason for your answer. [2 marks]
- (ii) In which country will there be a **greater** displacement of native species? Give a reason for your answer. [2 marks]
- (b) Outline ONE way in which a change in lifestyle can mitigate the impacts identified in Table 2. [3 marks]
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- (c) Define the term 'sustainable development'. [2 marks]
- (d) Explain how high fertility rates may impact on a country's ability to achieve sustainable development. [6 marks]
- (e) (i) Suggest an appropriate fertility rate that would allow Caribbean countries to achieve sustainable development. [1 mark]
- (ii) Justify your response in (e) (i) above. [4 mark]

Total 20 marks

MODULE 3

Answer BOTH questions.

5. Figure 3 shows variation in tree density from 1980 to 2000 in three Caribbean countries. Study Figure 3 and answer the questions that follow.

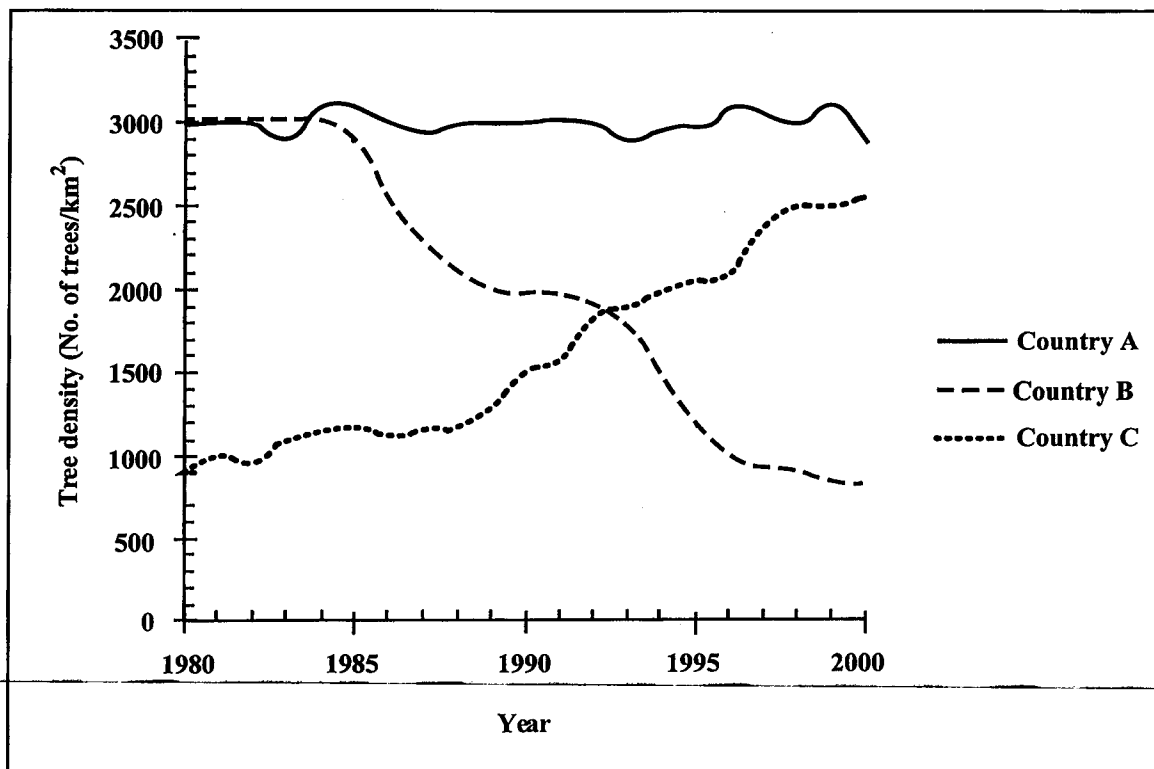


Figure 3. Variation in the tree density from 1980 to 2000 in three Caribbean countries

- (a) (i) Identify the country, A, B or C, that practises consumptive use of trees. [1 mark]
- (ii) Explain the reason for your answer in (a) (i) above. [2 marks]
- (iii) In what year did the country you identified in (a) (i) above begin to practise consumptive use? [1 mark]
- (iv) During the decade 1985 – 1995, what rate of change of tree density was observed in the country identified in (a) (i) above? Show your working. [3 marks]
- (v) Which country demonstrated the LEAST change in tree density? [1 mark]

- (b) (i) Other than logging, list THREE examples of consumptive use of natural resources important to the Caribbean. **[3 marks]**
- (ii) A country wishes to invest in tourism as a non-consumptive use of their beautiful beaches. Several large hotels are planned for one of the country's best beaches. Explain THREE possible environmental impacts of this proposed development on the area. **[6 marks]**
- (iii) To encourage natural resource conservation, the country in (b) (ii) above implements the following measures and tools for conservation:
1. Sustainable yield management
 2. Use of substitutes for non-renewable resources
 3. Reduction and minimization of waste
- Which ONE of the three measures listed above would apply to the proposed development in the country in (b) (ii)? **[1 mark]**
- (iv) Explain why the other measures would NOT apply to the proposed development in the country. **[2 marks]**

Total 20 marks

6. (a) (i) Using ONLY the examples of natural resources provided below, copy and complete Table 3 by inserting TWO renewable and TWO non-renewable natural resources.

Natural resources:

- Ground water
- Bauxite
- Seagrass beds
- Sand
- Oil
- Gravel
- Beaches
- Coral reefs
- Gold
- Soil

TABLE 3: CLASSIFICATION OF DIFFERENT TYPES OF NATURAL RESOURCES

Type of Natural Resource	
Renewable	Non-renewable
1.	1.
2.	2.

[4 marks]

- (ii) Discuss the effect of geographical and technological factors on the use of natural resources. Use suitable examples to illustrate your response. [6 marks]

- (b) Table 4 shows data on the growth in population and gravel extraction activities in a country from 1900 to 2000. Study the data and answer the questions that follow.

**TABLE 4: POPULATION GROWTH AND GRAVEL EXTRACTION,
1900 TO 2000**

Year	Population (millions)	Gravel extracted (000's of tonnes)
1900	0.69	0.15
1920	0.72	0.59
1940	0.87	0.87
1960	0.95	1.66
1980	1.02	2.86
2000	1.31	4.03

- (i) Describe the patterns in population growth and gravel extraction in the period 1900 – 2000. **[2 marks]**
- (ii) Explain the relationship between population growth and gravel extraction illustrated in Table 4. **[2 marks]**
- (iii) Estimate the amount of gravel that would be extracted in the year 2020. **[2 marks]**
- (iv) Gravel extraction can have significant negative impacts on the environment. Suggest TWO ways in which these impacts can be mitigated. **[4 marks]**

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.